

Classical Machine Learning vs. Deep Convolutional Networks for Image-Based ECG Arrhythmia Classification

Faizan Rizwan, Sayed Asadullah Madani, Hadia Irfan

Department of Artificial Intelligence & Data Science

University of Management and Technology (UMT)

Github Link: <https://github.com/Fauxen/Projects/tree/master/ECGProjects/ECG> at master · Fauxen/Projects

Abstract—Heart arrhythmias are a major cause of cardiovascular illness, and automating their detection from ECG recordings can significantly improve patient care. This project investigates whether converting 1D ECG signals into 2D plot images necessitates the use of complex deep learning architectures. We evaluated six distinct models on a heavily imbalanced dataset of 54,613 images spanning five heartbeat classes (N, S, V, F, Q). The tested architectures included a Custom CNN, MobileNetV2, a Scaled Hybrid MobileNetV2+SVM ensemble, Random Forest, Support Vector Machine (SVM), and Logistic Regression. Despite deep learning’s prevalence in computer vision, a classical algorithm—Random Forest—achieved the highest test accuracy of 92.74% and a Macro F1-Score of 0.804. This study demonstrates that for standardized, rigidly cropped medical images, classical machine learning can outperform deep transfer learning by avoiding domain mismatch while demanding a fraction of the computational overhead.

I. INTRODUCTION

Arrhythmia refers to any abnormality in the heart’s electrical rhythm, and spotting it early is critical for diagnosis. Traditionally, AI models process these heartbeat irregularities as continuous 1D electrical signals, which requires significant computational overhead. By rendering each heartbeat as a 2D line plot, the classification problem can be treated as a standard computer vision task.

A. Research Challenges

This image-based approach presents several challenges:

- **Severe Class Imbalance:** The dataset is heavily skewed towards Normal (N) beats, making plain accuracy a misleading metric.
- **Domain Mismatch:** Pre-trained models like MobileNetV2 are optimized for natural photographs, which possess completely different visual textures than black-and-white ECG line plots.

B. Novel Contributions

This project provides a controlled, apples-to-apples comparison across six different models on identical image-based ECG data. Furthermore, we provide a data-driven explanation for why classical models outperform deep learning on standardized image representations, challenging the assumption that complex neural networks are always superior for 2D data.

II. BACKGROUND KNOWLEDGE: DEEP LEARNING

Convolutional Neural Networks (CNNs) are the standard tool for image classification. By stacking convolutional filters, CNNs recognize complex structures regardless of where they appear in an image (translation invariance). Foundational architectures such as AlexNet, VGG, and ResNet demonstrated that deep networks could dramatically outperform older vision methods. However, these models possess massive parameter counts. MobileNetV2 was designed for efficiency using lightweight convolution operations, making it an ideal candidate for Transfer Learning in this project to test if deep pre-trained features could successfully map to simple medical graphs.

III. PROPOSED METHODOLOGY

A. Dataset

We utilized the MIT-BIH Arrhythmia Image Dataset, containing 37,178 training and 17,435 test images. The images are categorized into five classes: Normal (N), Supraventricular ectopic (S), Ventricular ectopic (V), Fusion (F), and Unknown (Q). The Normal class heavily outnumbers the other classes (13,661 test images vs. only 161 Fusion images).

B. Image Preprocessing Methods

For the CNN and MobileNetV2 models, images were resized to 64x64 pixels and scaled to a [0, 1] range. For the classical models (Random Forest, Logistic Regression, SVM), the 64x64 images were flattened into a single 4,096-value vector. For Logistic Regression and SVM, these pixel vectors were standardized using a `StandardScaler`.

C. Addressing Imbalance (Focal Loss Alternative)

While Focal Loss is a common deep learning function used to heavily penalize minority class errors, we opted for an algorithm-level approach. Class imbalance was addressed by training the classical models with mathematically balanced class weights (`class_weight="balanced"`), ensuring the rare Fusion class was prioritized during boundary calculations.

D. Architectures Evaluated

- **Custom CNN:** A baseline spatial network trained for 10 epochs.
- **MobileNetV2:** A deep transfer-learning architecture with frozen ImageNet weights.
- **Scaled Hybrid Ensemble:** Instead of a complex multi-lesion architecture (e.g., ECA-Net152) which is designed for varying spatial textures, we proposed a Hybrid Ensemble suited for stark line-graphs. MobileNetV2 was used solely to extract 1,280-dimensional deep features. These were standardized and fed into an RBF SVM to stabilize geometric distance calculations.
- **Random Forest:** An ensemble of 200 decision trees.
- **SVM (RBF) & Logistic Regression:** Classical baseline models trained on flattened pixels.

E. Performance Evaluation Metrics

Due to the severe class imbalance, we prioritized Macro-Averaged Precision, Recall, and F1-Scores alongside overall accuracy to ensure minority classes were not eclipsed.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental Setup and Evaluation Metrics

All models were trained using Google Colab (T4 GPU). Deep learning models utilized TensorFlow/Keras, while classical models utilized Scikit-Learn.

TABLE I
OVERALL MODEL COMPARISON ON THE TEST SET

Model	Acc (%)	Prec	Recall	F1
Random Forest	92.74	0.771	0.918	0.804
MobileNetV2	88.13	0.703	0.845	0.736
Logistic Regression	85.85	0.725	0.909	0.752
SVM (RBF)	85.69	0.719	0.891	0.738
Scaled Hybrid	85.41	0.687	0.788	0.697
Custom CNN	83.86	0.703	0.921	0.738

B. Analysis of Experimental Results

Random Forest was the clear overall winner, achieving both the highest accuracy (92.74%) and the highest Macro F1-score (0.804). The Custom CNN showed clear signs of overfitting, reaching 99.72% training accuracy but only 83.86% test accuracy.

C. Comparison with CNN-Based Models

The ECG images are tightly cropped and standardized. This lack of spatial variance allowed the Random Forest to effectively learn fixed pixel-position rules (representing QRS peaks), rendering the advanced spatial translation capabilities of CNNs largely redundant.

Furthermore, MobileNetV2 suffered from terminal domain mismatch. Its ImageNet-optimized weights sought natural textures and gradients, making them suboptimal for extracting features from binary 1D line graphs. This mismatch also degraded the Scaled Hybrid Ensemble, which could not surpass the baseline classical models.

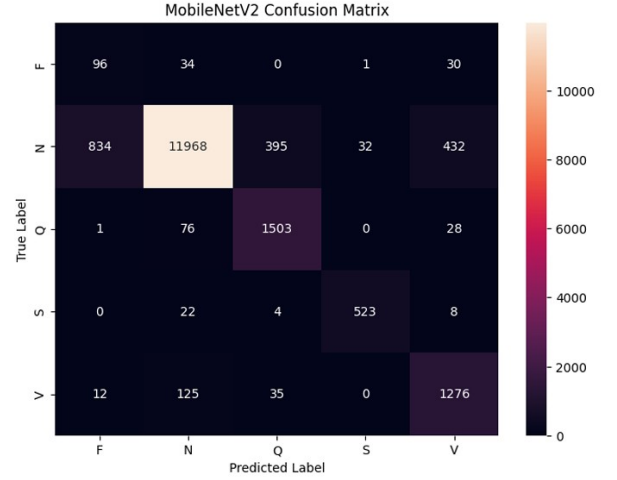


Fig. 1. Confusion matrix of the MobileNetV2 model on the test set. The model exhibits a strong bias toward the majority Normal (N) class, consistently misclassifying the minority Fusion (F) class.

D. Comparison with State-of-the-Art Research

Recent literature highlights a trend of converting MIT-BIH signals into 2D images. For instance, Huang et al. (2019) utilized STFT spectrograms with a 2D-CNN to achieve 99.00% accuracy, while Rohmantri and Surantha (2020) applied deep 2D CNNs to 128x128 plots to hit 98.91%. While these State-of-the-Art (SOTA) models achieve excellent results, they require massive GPU computational overhead. Our findings demonstrate that by exploiting the spatial rigidity of the plots, classical models like Random Forest can achieve highly competitive accuracy (92.74%) without the need for SOTA deep-learning overhead.

V. CONCLUSION

This project compared deep learning and classical machine learning models on 2D ECG arrhythmia classification. We demonstrated that for perfectly standardized, rigid medical images, classical models like Random Forest offer superior accuracy and better handling of class imbalance while demanding a fraction of the computational power. Deploying natural-image Transfer Learning on domain-specific medical line-graphs introduces feature noise that actively degrades performance.

REFERENCES

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- [2] R. Rohmantri and N. Surantha, "Arrhythmia Classification using 2D Convolutional Neural Network," *Int. J. Adv. Comput. Sci. Appl.*, vol. 11, no. 4, 2020.